Academic Year 2025–2026

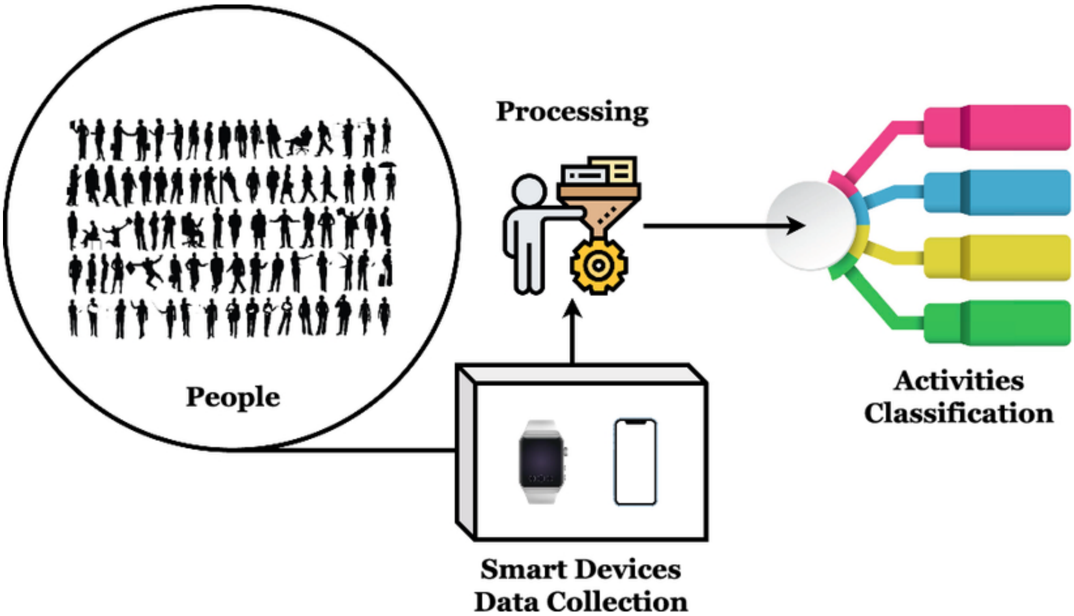
Adaptation of Time series forecasting Large language Model method to Human Activity Recognition (HAR) task

Project Type: [ ] "Research" [√] "Research and Development" [ ] "Development"

Supervisors: Wenda LIU, Fabien PICAROUGNE, Guillaume RASCHIA

Email: wenda.liu@etu.univ-nantes.fr

Research Background



Human Activity Recognition (HAR) is one of the core application areas in time series analysis. It infers human activity states (walking, running, sitting, etc.) through sequential data collected by sensors (such as acceleration and angular velocity), with wide applications in health monitoring, sports science, smart homes, and other scenarios. The WISDM (Wireless Sensor Data Mining) dataset, as a classic benchmark dataset in the HAR field, contains a large amount of time series data collected by wireless sensors, providing standardized support for algorithm performance verification.

In recent years, time series forecasting models based on Large Language Models (LLMs) have made significant progress. Among them, the **AutoTimes** model demonstrates excellent long-sequence modeling capabilities in various time series forecasting tasks through "time series-text modality conversion" and autoregressive LLM architecture. The core advantage of this model lies in converting continuous time series into embeddings that are understandable by LLMs, leveraging generalization capabilities of pre-trained LLMs to achieve high-precision forecasting. However, the AutoTimes model is mainly designed for "forecasting" downstream tasks (such as predicting sensor values at future moments) and has not been adapted for "classification" tasks in the HAR field (such as judging activity categories based on sensor time series data). Therefore, how to preserve the time series encoding advantages of the AutoTimes model while achieving HAR classification adaptation through downstream task head transformation, and verifying its effectiveness on the WISDM dataset, has become a key issue to be solved.

Research Objectives

The core objective of this project is to migrate the **AutoTimes** model from time series forecasting task to time series classification task on **WISDM dataset**, and complete performance evaluation and optimization, including:

1. Reproduce the original forecasting task performance of the AutoTimes model on the WISDM dataset to establish a basic performance benchmark;
2. Transform the downstream task layer of the AutoTimes model: preserve the "time series-text encoding" module and LLM backbone network of the original model, design and train output heads suitable for HAR classification (such as fully connected classification layers);
3. Verify the classification performance of the transformed model on the WISDM dataset and compare it with traditional time series classification methods (such as FCN);
4. Analyze the impact of AutoTimes model's modality conversion strategy and LLM feature extraction capabilities on classification performance.

Work to be Completed

1. **Literature Research and Organization**:
    - Read related work to build knowledge of time series classification, time series forecasting, AutoTimes model, and HAR tasks.
2. **Data Preprocessing and Benchmark Construction**:
    - Clean the WISDM dataset (handling missing values, outliers), standardize (such as sequence length alignment, feature normalization), and split (training/validation/test sets);
    - Reproduce the forecasting task of the AutoTimes model on the WISDM dataset (such as predicting acceleration values for the next 10 moments), record MAE, RMSE and other metrics to establish the basic performance benchmark of the model;
3. **AutoTimes Model Classification Task Transformation**:

- Fix the parameters of the "time series-text encoding" module and LLM backbone network of AutoTimes, only design and train classification task heads (such as global average pooling + fully connected layers based on LLM output features, outputting probability distributions for activity categories; freely using out-of-shelf model and AI assitant);
- Optimize training strategies: design loss functions for classification tasks (such as cross-entropy loss), adjust learning rates and batch sizes to avoid overfitting of classification heads;

4. **Performance Evaluation and Comparative Analysis**:
    - Test the classification performance of the transformed model on the WISDM test set, record accuracy, F1-score, confusion matrix and other metrics;
    - Compare with traditional time series classification methods (FCN)
5. **Result Analysis and Report Writing**:
    - Analyze the impact of modality conversion methods, LLM feature dimensions, and classification head structures on performance, summarize the advantages and disadvantages of the model, and propose future optimization directions (such as adding local feature attention, multimodal fusion).

Required Skills

- **Programming Languages and Tools**: Python (Pandas, NumPy for data processing, PyTorch for model implementation);
- **Machine Learning Fundamentals**: Understand Deep learning principles;

References

[1] Wang, Zhiguang, Weizhong Yan, and Tim Oates. "Time series classification from scratch with deep neural networks: A strong baseline." 2017 International joint conference on neural networks (IJCNN). IEEE, 2017.
[2] Flunkert, Valentin, David Salinas, and Jan Gasthaus. "DeepAR: Probabilistic forecasting with autoregressive recurrent networks." arXiv preprint arXiv:1704.04110 23 (2017).
[3] Liu, Chenxi, et al. "Towards Cross-Modality Modeling for Time Series Analytics: A Survey in the LLM Era." arXiv preprint arXiv:2505.02583 (2025).
[4] Liu, Y., et al. "AutoTimes: Autoregressive time series forecasters via large language models." arXiv preprint arXiv:2402.02370 (2024).
[5] WISDM Dataset. https://www.cis.fordham.edu/wisdm/dataset.php